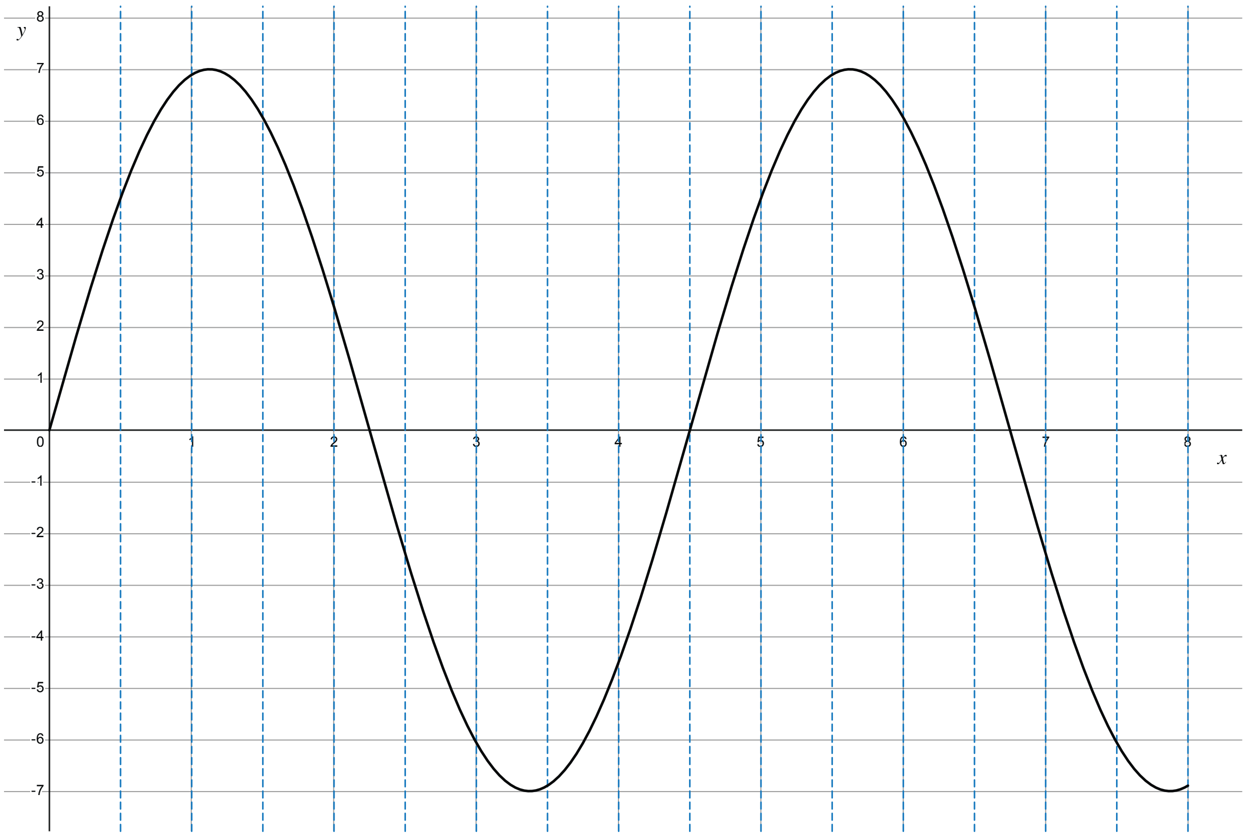

4. Activity: Sample the Wave

Use the printed graph handout for this activity. The graph shows a signal plotted from t = 0 to t = 8, with dashed vertical lines at every 0.5 seconds and integer gridlines on the y-axis (−7 to +7).

For each sample: follow the dashed vertical line to where it meets the curve, read the y-value (voltage), round to the nearest integer, then encode in 4-bit sign-magnitude binary using the reference table in §3.3.

Part A — Fine sampling

Sample at **t = 1, 2, 3, 4, 5, 6, 7, 8**.

Sample	Time (s)	Voltage (from graph)	Rounded	Binary
1	1			
2	2			
3	3			
4	4			
5	5			
6	6			
7	7			
8	8			

Part B — Coarse reconstruction

Imagine the ADC only recorded samples at **t = 2, 4, 6, 8** — one sample every 2 seconds instead of every 1 second. You already have those four values from Part A.

On the printed graph, mark those four points and connect them with straight line segments. This is what a DAC would reconstruct if that coarse data was all it had.

Discussion:

- Where does the reconstruction go most wrong compared to the original wave?
- What features of the signal did the lower sample rate fail to capture?

Value	Sign bit	Value bits	Full binary
+7	0	111	`0111`
+6	0	110	`0110`
+5	0	101	`0101`
+4	0	100	`0100`
+3	0	011	`0011`
+2	0	010	`0010`
+1	0	001	`0001`
0	0	000	`0000`
−1	1	001	`1001`
−2	1	010	`1010`
−3	1	011	`1011`
−4	1	100	`1100`
−5	1	101	`1101`
−6	1	110	`1110`
−7	1	111	`1111`